Signal Sport: CapLap

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INTRODUCTION

The modern NBA is shaped not only by player talent, but also by salary cap rules, luxury tax penalties, and apron restrictions that affect how teams build their rosters. However, this information can be difficult to follow because it is often presented in complicated articles, spreadsheets, or rule-heavy explanations. Users may understand that cap space matters, but not always how it influences trades, free agency, or long-term roster flexibility. Signal Sports is a proposed web app designed to make NBA cap and roster information easier to understand through a more visual and interactive interface.

OBJECTIVE

NBA salary cap rules, luxury tax penalties, and apron restrictions play a major role in roster construction, yet this information is often difficult to understand because it is spread across articles, spreadsheets, and rule-heavy explanations. This can make it harder for users to evaluate team payroll, roster flexibility, and the financial impact of trades or free agency decisions. The objective of this project is to develop a web application that presents NBA cap and roster information in a clearer, more visual, and interactive way for coaches, analysts, content creators, and fans.

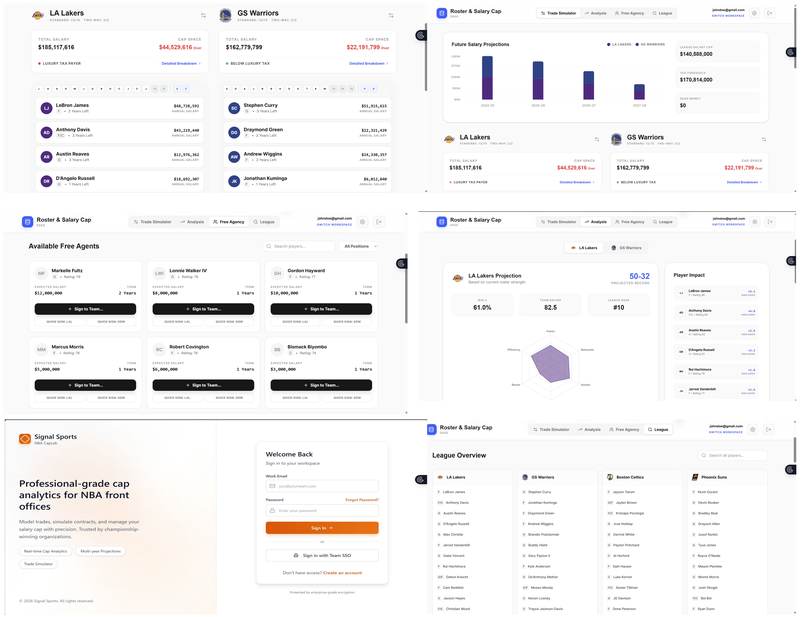
DEVELOPMENT PROCESS

- Began by exploring possible app names and overall project direction to define a clear identity for the platform.
- Initially considered combining both NBA and NFL team-building analysis into one application but later narrowed the focus to the NBA after recognizing that each league has different roster rules, financial systems, and user needs. The NBA was selected as the starting point because of stronger personal familiarity with the league.
- Developed early prototype ideas using both Figma and Google AI Studio to compare interface concepts and identify the strongest design elements from each version.
- Considered future feature expansions, including team analytics tools to evaluate how trades and roster changes could affect projected wins and team performance, as well as AI-driven features to provide suggestions and support user decision-making.
- Originally defined the target users as coaches, front-office personnel, and scouting staff, but later broadened the audience after feedback suggested making the experience more game-like and engaging for fans as well.
- Still in the design and prototyping stage, with full development not yet started because the project requires additional backend knowledge, particularly in areas related to data handling and user privacy.
- Plan to continue research on backend development and begin building the full application during the summer after the current academic term ends.

TOOLS/TECHNOLOGIES

- **Figma and Google AI Studio:** prototype design, interface exploration, and concept development
- **React.js, HTML, CSS, and JavaScript:** frontend web development
- **AWS Amplify and Amazon Cognito:** web app hosting and user authentication
- **Amazon RDS/Amazon Aurora:** relational database and data management
- **Adobe Photoshop and Illustrator:** visual assets, interface elements, and logo design
- **GitHub:** version control and project management

PROTOTYPE FEATURES



CONCLUSION

Signal Sports: CapLap presents a proposed web application designed to make NBA salary cap, roster, and team-building information easier to understand through a more visual and interactive interface. By addressing the complexity of cap rules, tax penalties, and roster flexibility, the project aims to support clearer decision-making for coaches, analysts, content creators, and fans. While the platform is still in the design and prototyping stage, it establishes a strong foundation for future development into a fully functional web application.